

# Carson Manuel Montini

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## Education

**California Baptist University, Riverside, CA** Expected Apr 2027  
**Bachelor of Science in Mechanical Engineering** | Physics Minor | ABET-accredited program

## Technical Skills

**CAD and design:** SolidWorks, Fusion 360, AutoCAD, design for manufacturing, mechanical packaging  
**Analysis:** Finite element analysis (FEA), computational fluid dynamics (CFD), MATLAB/Simulink, structural testing  
**Manufacturing:** CNC machining, CAM, mills, lathes, welding, additive manufacturing, carbon-fiber fabrication  
**Programming and tools:** C++, Arduino, Microsoft Excel, Microsoft Office | Spanish biliteracy

## Engineering Experience

**Battery Mechanical Design Engineer | Formula SAE Electric Team** Sep 2023 - Present

*California Baptist University, Riverside, CA*

- Designed the accumulator enclosure, segment packaging, connector interfaces, and precision components in SolidWorks for safe, serviceable vehicle integration.
- Reduced battery weight 12% and accumulator volume 20% through structural and packaging optimization while preserving safety and manufacturability requirements.
- Designed and evaluated nickel busbar fuse geometries using fuse-width calculations, transient thermal analysis, and infrared testing to characterize current-interruption behavior.
- Coordinated mechanical and electrical interfaces across 10 vehicle subsystems and prepared component designs for CNC manufacturing using design-for-manufacturing practices.

## Selected Engineering Projects

**Automated Court Enhancer | Multidisciplinary Team Project** Nov 2025 - Apr 2026

- Designed and assembled a 12-inch autonomous hardwood-court cleaning robot integrating dual DC drive motors, a rotating cleaning pad, fluid pump and misters, and onboard water storage.
- Sized mechanical and electrical components against 1.0-1.2 m/s speed and 2-hour runtime design targets; developed 12 V power distribution and 5 V control architecture.
- Integrated Arduino Mega and H-bridge controls to coordinate drive, cleaning, and fluid-delivery functions within a compact chassis.

**Reverse-Engineered Industrial Clamp | Additive Manufacturing Team Project** Sep 2023 - Nov 2023

- Reverse-engineered an industrial clamp in SolidWorks and developed a finite element model to evaluate stress distribution, deformation, and critical load paths.
- Applied topology optimization to remove low-stress material while preserving stiffness and strength requirements, then converted the result into a manufacturable bracket design.
- Conducted mesh-refinement and convergence studies to verify solution stability, then compared simulation trends with three-point bending and tensile-test results from Markforged carbon-fiber-reinforced specimens.

## Leadership Experience

**Resident Advisor** Aug 2024 - Apr 2026

*California Baptist University, Riverside, CA*

- Mentored residents, planned community programs, communicated university policies, and responded to conduct and safety concerns in a residential setting.

**Living Area Coordinator** Sep 2023 - Apr 2024

- Planned weekly service activities and coordinated community events to strengthen resident participation and engagement.